

PMA 804C - calculated airflow and pressure capability

Purchased configuration: 7.5 kW motor, maximum 1600 rpm. Base performance source: supplied PMA 804C datasheet at 5.5 kW / 1400 rpm.

Axial fan VFD controlled Ø800 mm nominal

Confirmed base data from supplied datasheet

Parameter	Published PMA 804C data
Maximum airflow	38,200 m ³ /h
Impeller speed	1,400 rpm
Motor power	5.5 kW
Voltage	400/690 V - D/Y - 50 Hz
Current	10.8 A
IP rating	IP55
Weight	106 kg
Medium temperature	-20 to +40 °C
Dimensions	ØC int 810; ØD 800; E 560; ØF 855; ØG 884; 12 holes Ø14

Purchased execution and calculation basis

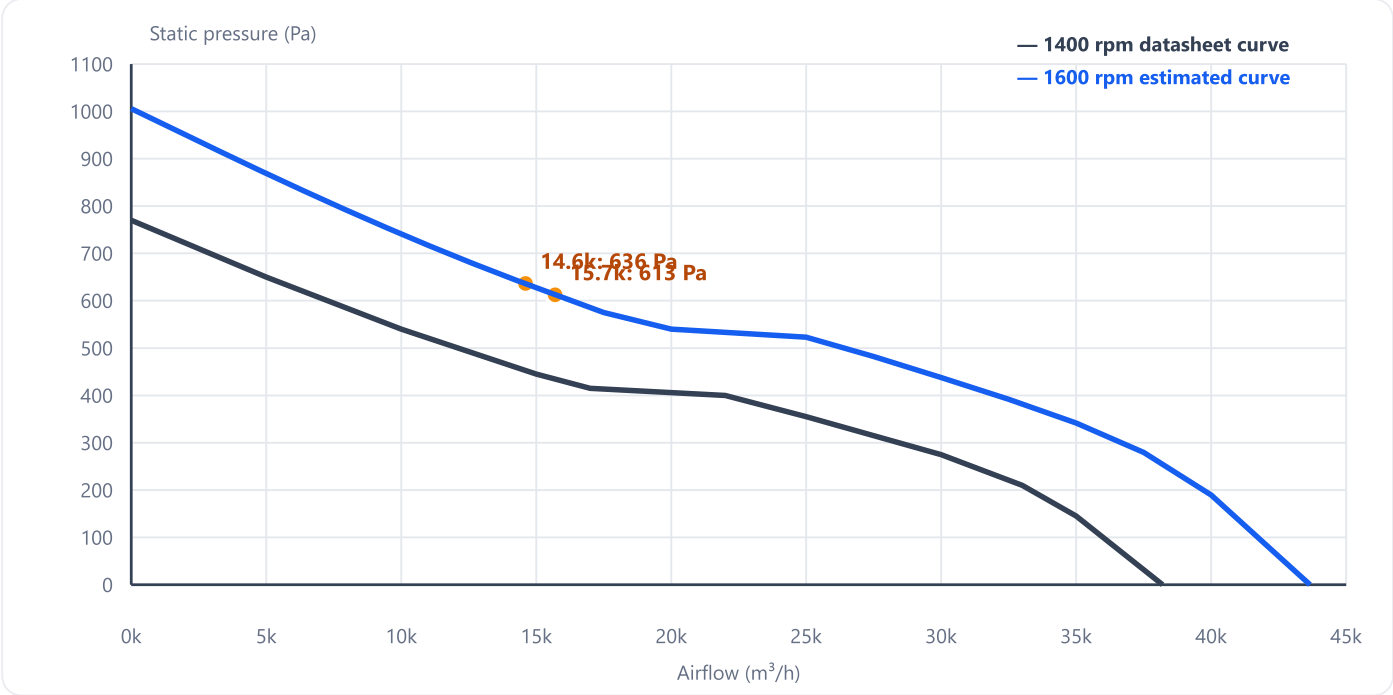
Important: the attached manufacturer sheet is for the standard 5.5 kW / 1400 rpm version. The installed/purchased fan is user-confirmed as **7.5 kW with 1600 rpm maximum**. Values at 1600 rpm below are therefore engineering estimates using fan affinity laws, not a published BVN/Slingerland test curve.

Speed ratio 1.143	Estimated free-air flow at 1600 rpm 43,657 m³/h	Pressure scaling factor 1.306x
Power scaling factor		

1.493x

Affinity laws used: $Q \propto n$, $\Delta p \propto n^2$, $P \propto n^3$. The 7.5 kW motor can therefore become the limiting factor before the theoretical 1600 rpm curve is reached at every operating point.

Approximate static-pressure capability



The 1400 rpm black curve was digitized approximately from the supplied graph. The blue curve is calculated from it by affinity-law scaling.

Airflow	Approx. static pressure @ 1400 rpm	Estimated static pressure @ 1600 rpm
10,000 m³/h	540 Pa	741 Pa
12,500 m³/h	492 Pa	682 Pa
14,600 m³/h	453 Pa	636 Pa
15,700 m³/h	434 Pa	613 Pa
20,000 m³/h	406 Pa	540 Pa

25,000
m³/h

355 Pa

523 Pa

Implication for the process-air system

At the filter design flow of **14,600 m³/h**, the extrapolated 1600 rpm fan curve is approximately **636 Pa static pressure**. At **15,700 m³/h** it is approximately **613 Pa**.

If the roll filter is allowed to reach its suggested maximum pressure drop of **150 Pa**, the remaining static-pressure budget at 14,600 m³/h is approximately **486 Pa** before allowing for the recuperator, ducts, bends, transitions, roof outlet and design margin.

This is a system-design estimate. Final VFD maximum frequency / rpm should be commissioned against actual motor current, absorbed power and measured system pressure.

Source: supplied PMA 804C datasheet (5.5 kW / 1400 rpm). Purchased 7.5 kW / 1600 rpm execution supplied by project owner. Calculated values are clearly separated from manufacturer-published values.